

docs-ari-strategy

ARI Adoption Strategy — How a spec becomes a standard

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ARI Adoption Strategy — How a spec becomes a standard

This document is the playbook for turning **ARI** (`../ARI-SPEC.md`) from a doc in one repo into an interface the industry targets. It is modeled on how durable infrastructure standards actually won (OpenStack, the Open Compute Project, POSIX, JDBC, the Language Server Protocol) rather than on how products launch.

The thesis in one line: a standard wins by claiming a layer the incumbents *structurally cannot follow you into*, proving it with a reference implementation you dogfood, and making conformance *testable* so others can credibly say "we implement it."

1. Why this can work (the structural bet)

The mature agent frameworks — LangGraph, CrewAI, AutoGen — are Python-locked by design. That is fine for server apps, but it means they **cannot** serve:

- embodied / robotic systems (real-time, no GC pauses),
- edge / on-device deployments (constrained, often offline, local models),
- agents embedded inside native apps (game engines, DSP, trading, CAD).

ARI's **ARI-Embedded** profile is defined precisely around requirements those runtimes cannot meet without ceasing to be Python-locked (native core, no hot-path allocation, bounded memory). That is the moat. We are not trying to be "a faster LangGraph" on LangGraph's turf — a losing standards fight. We are defining the lane they can't enter.

This mirrors the Cole Crawford pattern: pick the layer defined by a hard constraint (for him, the physics of edge latency/location; for us, the runtime constraints of embedded/embedded AI), then own both the **standard** and its **reference implementation**.

2. The three artifacts a standard needs

A standard is not a marketing claim; it is three things that must all exist:

Artifact	Status	Where
The spec — normative contract with MUST/SHOULD/MAY	drafted	ARI-SPEC.md
A reference implementation — proof the spec is buildable	exists	marrow (this repo)
A conformance kit — lets a <i>third party</i> prove they comply	exists (kit v0.1)	ari-conformance/

The third is what most "standards" are missing and why they stay vanity docs. Browsers got real because of the Acid tests; OpenStack interop got real because of RefStack. **Build the conformance kit early** — it is the thing that lets someone else say "ARI-Core + ARI-Embedded conformant" with evidence, and it is what makes the spec falsifiable rather than aspirational.

Conformance kit, concretely

- A language-neutral test corpus: JSON fixtures + expected behaviors for each numbered requirement in the spec (the [Appendix B checklist](#) becomes executable).
- A thin per-language harness that drives a runtime through the corpus.
- A published results format ("ARI 0.1 · Core+Embedded · kit v0.1 · pass 47/47").
- Run marrow against it in CI so the reference implementation is provably conformant and regressions are caught.

Status (2026-05-28): the kit now exists at [ari-conformance/](#) (kit v0.1, tests mapped to the [Appendix B checklist](#)) and runs in CI as the conformance job — now a **hard gate** (since commit 4302bd3, 2026-05-28, after its first confirmed green run on Linux). That flip is the move that makes "ARI-conformant" a falsifiable claim rather than a slogan.

3. The adoption ladder (dogfood → niche → ecosystem)

Do not chase external adopters first. Earn the right to be a standard by proving it on systems you control, then expand outward one credible step at a time.

Rung 1 — Dogfood in your own embodied system (you control the believer). Make ARI the runtime contract that **Tilo** (Unitree H2) actually runs on. This is the lighthouse deployment: a real humanoid running an ARI-Embedded runtime is worth more than any benchmark. It also forces the spec to be honest — anything that doesn't survive contact with a real robot gets fixed.

Rung 2 — Second independent proof point in your own portfolio. Embed an ARI runtime in a *latency-sensitive, non-robotic* native context — the trading stack is the obvious candidate. Two independent deployments across two domains, both yours, is the credibility floor before asking anyone external to care.

Rung 3 — A second implementation, not just a second user. A standard with one implementation is a library. Encourage (or write) a second conformant runtime — e.g. a Rust ARI-Core core — even a partial one. The moment two runtimes pass the same conformance kit, "interface" stops being a metaphor.

Rung 4 — Adapters that make ARI the path of least resistance. Ship adapters so ARI sits *with* the ecosystem, not against it:

- expose an ARI runtime's tools over **MCP**,

- let an ARI runtime participate in an **A2A** mesh,
- provide a shim so existing LangGraph/CrewAI tool definitions can be invoked through an ARI ToolRegistry. Being MCP/A2A-complementary ([spec §1.3](#)) means you ride the protocol wave instead of fighting it.

Rung 5 — Neutral governance. Once there is a second implementation and outside interest, move the spec out of this repo into a neutral home (its own ari-spec repo/org, CC BY 4.0 text, an open issue/RFC process). Standards die when they look like one company's property. Cole's standards (OpenStack, OCP) all moved to foundations precisely to shed that perception. Keep marrow as the reference implementation, but let the *spec* belong to everyone.

4. Positioning rules (how to talk about it)

- **Lead with embodied/edge, not speed.** "The agent runtime you can embed in a robot or edge device" — not "a faster LangGraph." Speed is a feature; the embedded lane is the identity.
 - **Always name the layer.** ARI is the *runtime* layer beneath MCP/A2A. Repeat the stack diagram everywhere. Owning a clear layer is how people know where to put you.
 - **Be honest about maturity.** The reference implementation is early; say so. Credibility compounds; overclaiming burns it once.
 - **Conformance language is disciplined.** Never "ARI-compatible" alone — always "ARI-Core" / "ARI-Embedded" + version. Sloppy conformance claims are how a standard's meaning erodes.
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5. Naming, trademark, and the spec/impl split

- **ARI** = the standard (the interface). **marrow** = the reference implementation. Keep these rigorously distinct in all materials — conflating them makes ARI look like a single vendor's API and kills neutral adoption.
 - ✓ **Naming collision resolved (2026-05-28): renamed agentcore → marrow.** The prior name collided head-on with **Amazon Bedrock AgentCore** (AWS; GA 2025-10-13), which ships an "AgentCore Runtime" targeting the *same layer* — it would have lost every search and invited trademark risk. The implementation is now **marrow** (PyPI: marrow-rt); **ARI** (the standard) is unchanged. That the standard survived an implementation rename is exactly why the two are kept distinct (above). Remaining founder tasks: rename the GitHub repo to marrow, claim the marrow-rt PyPI project, and grab a docs domain.
 - Reserve the obvious homes early (an ari-spec repo, a docs domain) so the standard has a neutral-looking address when Rung 5 arrives.
 - Spec text under **CC BY 4.0** (anyone can implement); reference code under **Apache-2.0** (patent grant, embeddable in proprietary products). This split is deliberate and is stated in [spec §13](#).
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6. The 90-day concrete next steps

1. **Merge the spec + repositioned README** so the repo reads as "reference implementation of ARI," not "another agent framework." (*done*)
2. **Stand up ari-conformance/** with the Appendix B checklist as executable tests; wired into CI against marrow (conformance job),

now a **hard gate** after its first confirmed green run on Linux (commit 4302bd3).

3. **Port Tilo's agent loop onto an ARI-Embedded build** of marrow; capture the gaps the real robot exposes and fold them into ARI 0.2.
4. **Write the MCP adapter** (expose ARI tools over MCP) — the cheapest way to be seen as ecosystem-complementary.
5. **Publish ARI 0.1 as an RFC** (issue + discussion), inviting one external reviewer from the embedded/robotics world.

The order matters: spec + reference impl + conformance kit + a real embodied deployment, *before* any push for outside adoption. That sequence is what turns "a doc I wrote" into "the interface other people target."